

Technology Stack

Date	03 October 2026
Team ID	SWTID-2026-5999
Project Name	PocketSmart AI — AI-Assisted Goal-Based Budget Planner
Maximum Marks	2 Marks

S.No	Architecture Component	Technology	Why It Fits the Project
1	Frontend	HTML5, CSS3, JavaScript, Jinja2	Lightweight interface without a separate frontend build system; suitable for a small academic web application
2	Backend	Python, FastAPI, Uvicorn	Fast API development, request validation and clear route structure
3	Data Validation	Pydantic	Keeps budgets, quantities, guest counts and text inputs within defined limits
4	AI Layer	Google Gemini through google-genai	Generates structured recommendations and can process the optional jewelry image
5	Database	SQLite	Simple relational storage for users and saved plans with no separate database server
6	Authentication	bcrypt and JWT	Secure password hashing and signed session tokens
7	Image Handling	Pillow	Validates and prepares uploaded images before multimodal processing
8	Testing	pytest, custom load-test script	Functional checks and basic concurrency/performance evidence
9	Version Control	Git and GitHub	Supports five-member collaboration, history and code review
10	Deployment Support	Docker	Provides a repeatable environment for later cloud deployment

Technology Selection Principle

The team deliberately uses a small, understandable stack. The project does not train its own machine-learning model; Gemini provides the generative component while the application’s backend remains responsible for validation, calculations, storage and access control.